

1.3 – DC Operating Point and Signal Variation

Analyzing Real Circuits

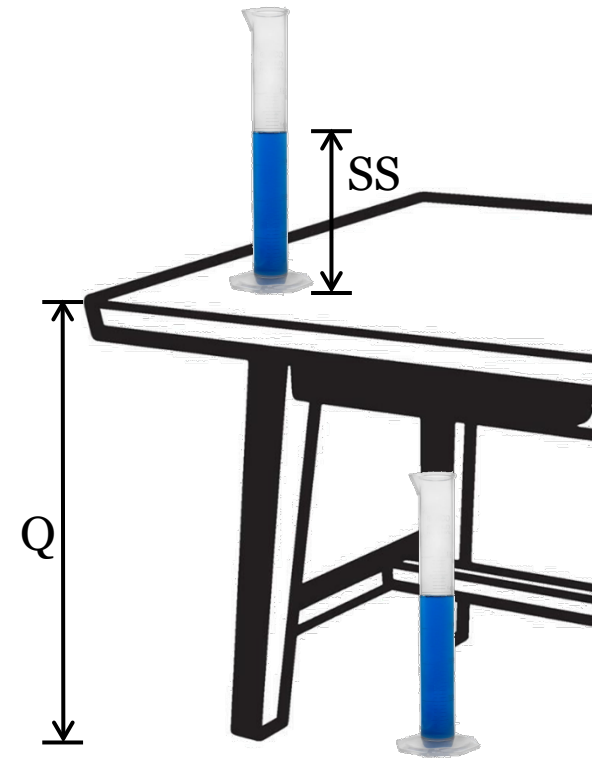
- Real circuits have:

- A DC Operating Point (Q)
- A Small-Signal (SS) response (linear only)
- Often neglect nonlinear AC effects

- Why?

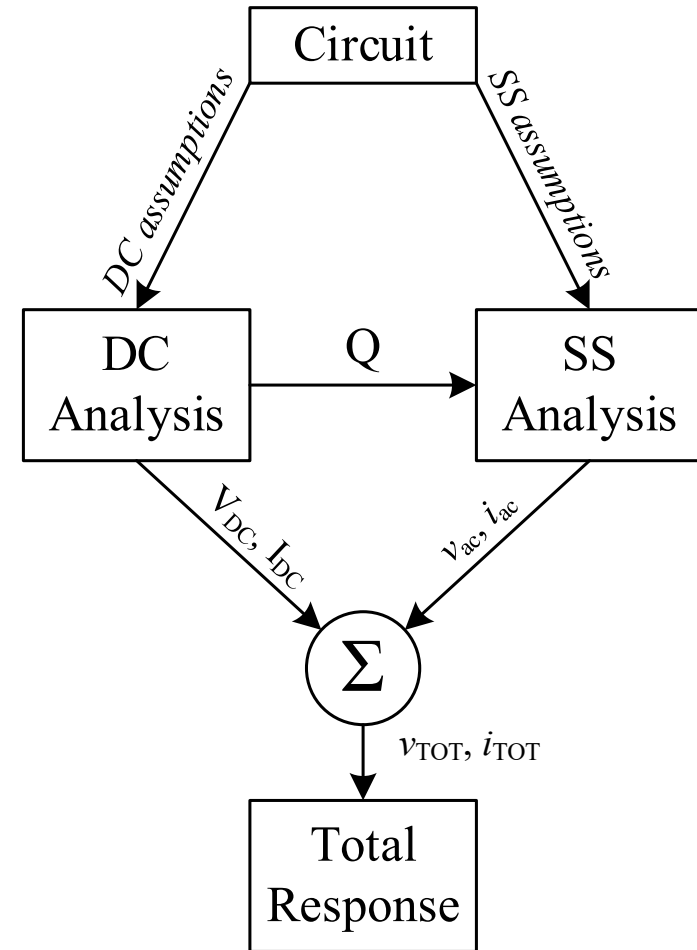
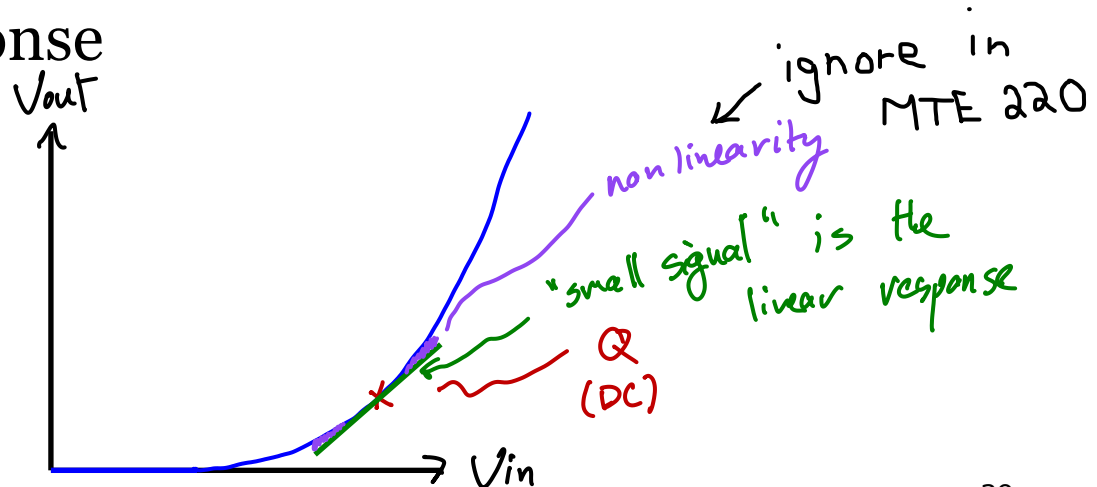
- To keep signals within the voltage rails
 - Example: A physics experiment (SS) on top of a table (Q) instead of the floor
- Sometimes the circuit response to SS depends on Q
 - Example: Turning on a gasoline engine (Q) then revving it (SS)

*"Quiescent" operating point
"Bias" voltages*



Decomposing Circuit Response

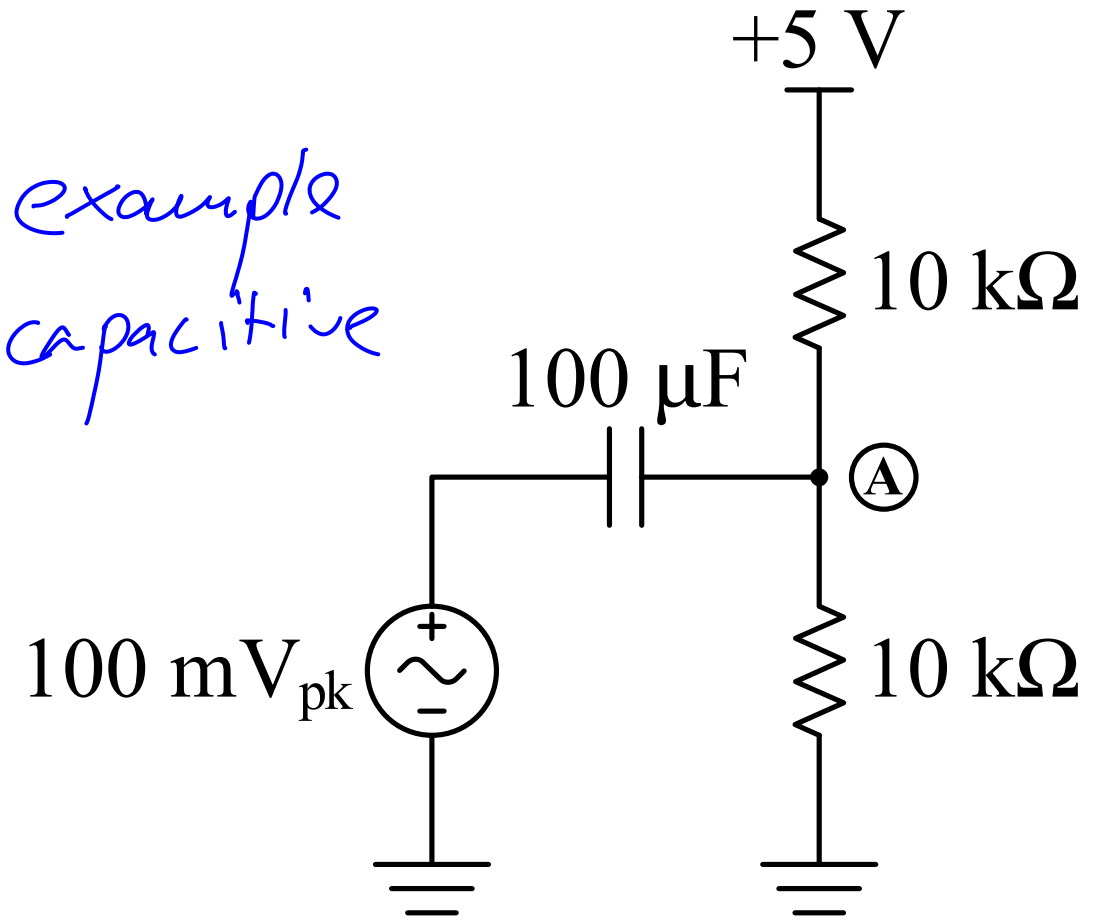
- Draw a DC version of the circuit
 - Determine Q
- Draw a SS version of the circuit
 - Determine SS response
- Sum the responses to get the total response



Example

- Determine v_A , the total response of this circuit

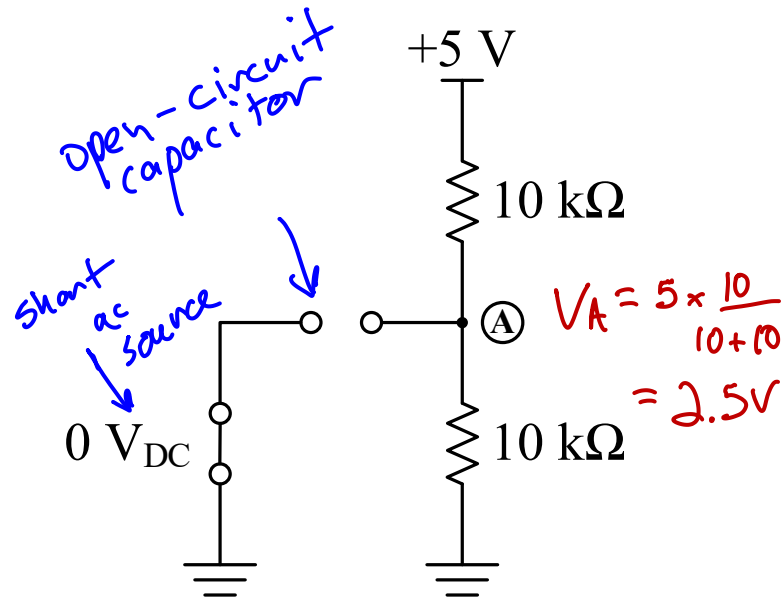
- This circuit is an example of level-shifting using capacitive coupling.



Redraw Circuit for DC and SS Analysis

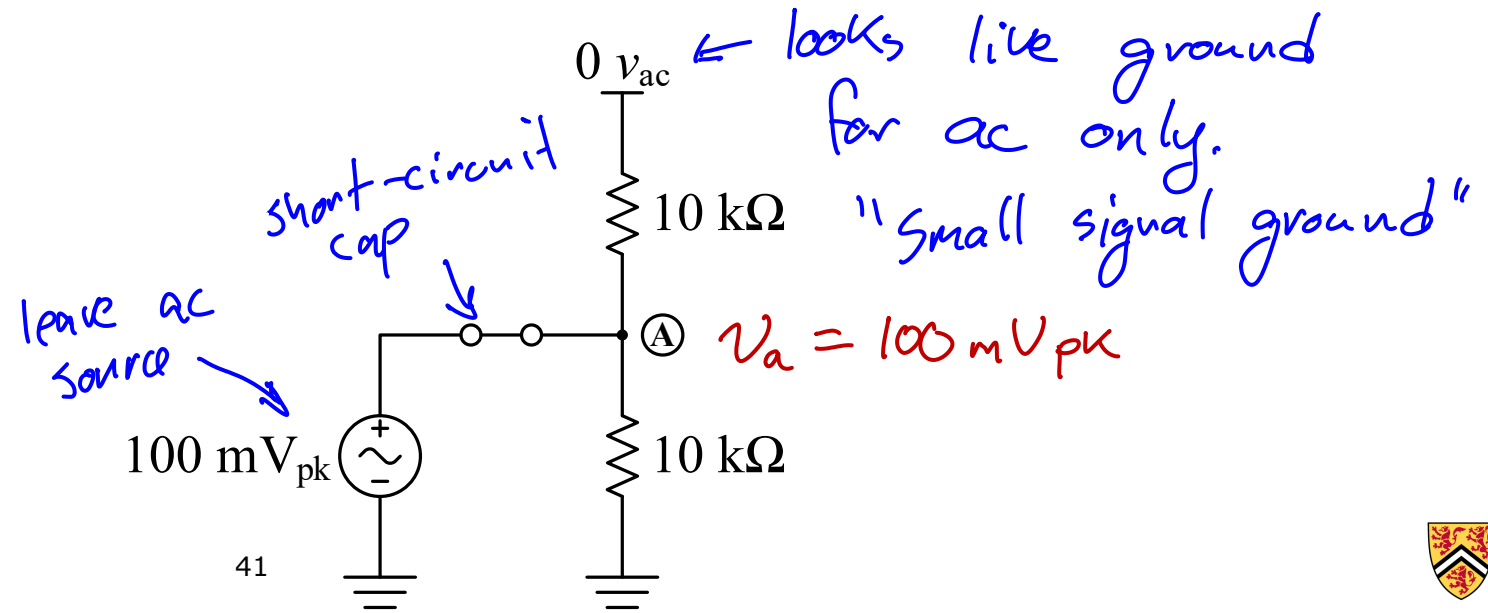
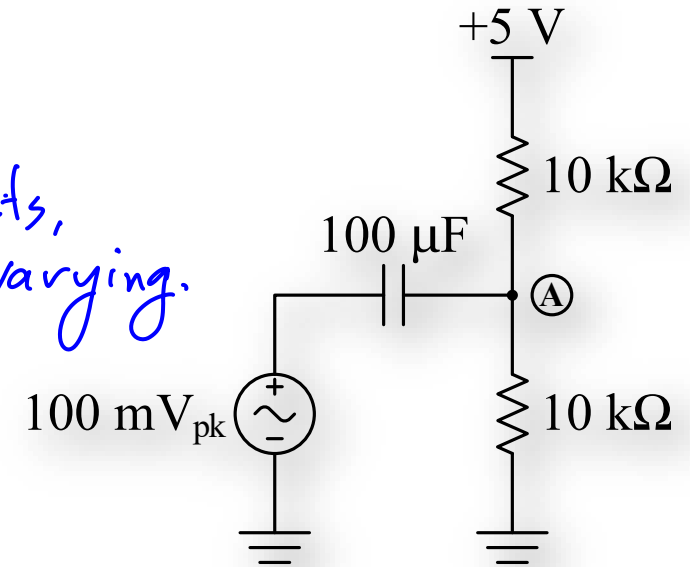
DC: *All derivatives are zero*

- $C \rightarrow \text{open}$
- $L \rightarrow \text{short}$
- $v_{ac} \rightarrow 0$ $V_{DC} \rightarrow \text{short}$
- $i_{ac} \rightarrow 0$ $I_{DC} \rightarrow \text{open}$

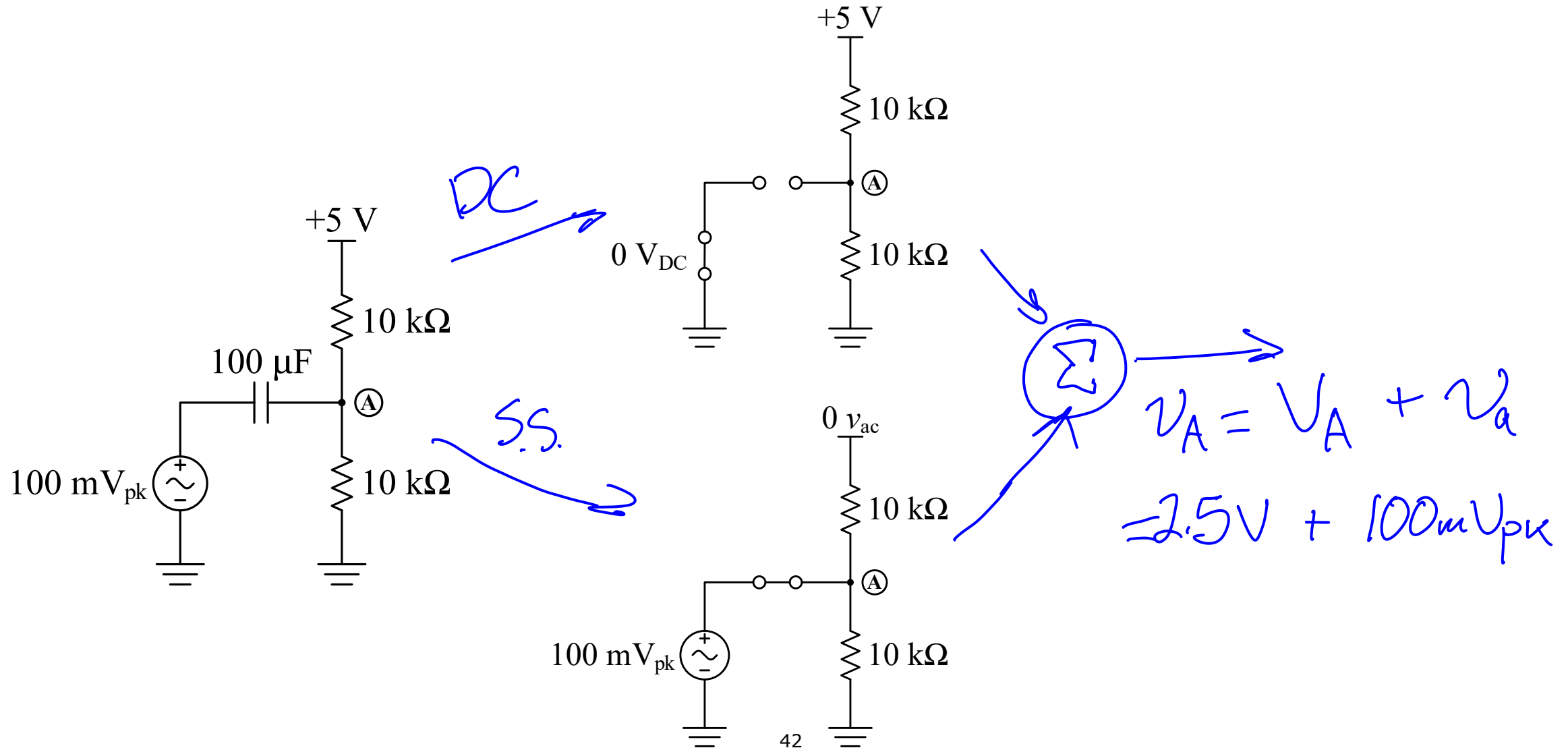


SS (AC): *No constants, only time-varying.*

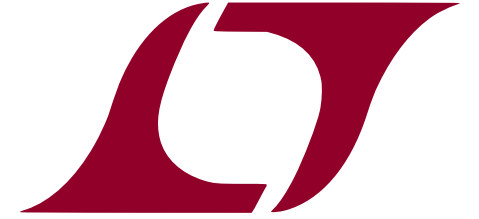
- $C \rightarrow \text{short}$
- $L \rightarrow \text{open}$
- $V_{DC} \rightarrow 0$ $v_{ac} \rightarrow \text{short}$
- $I_{DC} \rightarrow 0$ $i_{ac} \rightarrow \text{open}$



Sum to get Total Response



LTSpice Modeling Challenge



- Create the example circuit schematic
 - Perform a DC analysis (“DC op pnt”)
 - Perform a transient analysis (choose a frequency for your 100 mV source)
- How could you find only the AC response?
- What does the AC Analysis do, and how is that different?
- What happens if you change the value of C?

